

Scientific Computing Exercise Set 2

Jiawei Li(Student ID: 15988015)& Boyu Chen(Student ID: 16124510) & IJsbrand Meeter(Student ID: 13880624)

Repository: (*Scientific Computing*) https://github.com/ShioMura/Scientific_Computing

I. INTRODUCTION

Diffusion-driven processes are fundamental in physics, chemistry, and materials science. In the steady-state regime and in the absence of internal sources, diffusion is governed by Laplace's equation

$$\nabla^2 c = 0, \quad (1)$$

which describes harmonic scalar fields. When such a diffusive field is coupled to a moving boundary, complex growth patterns may emerge due to nonlinear feedback between geometry and flux.

The aim of this project is to implement a numerical solver for the two-dimensional Laplace equation using the Successive Over-Relaxation (SOR) method, to validate the solver under prescribed boundary conditions including an internal absorbing region, and to simulate diffusion-limited aggregation (DLA) using harmonic measure. Furthermore, the influence of a nonlinear growth parameter η on the morphology of the aggregate is investigated. This study demonstrates how deterministic diffusion governed by Laplace's equation, when coupled to stochastic boundary evolution, gives rise to nonlinear Laplacian instabilities and fractal growth.

II. THEORY

A. Laplace Equation

In two spatial dimensions, Laplace's equation reads

$$\nabla^2 c = \frac{\partial^2 c}{\partial x^2} + \frac{\partial^2 c}{\partial y^2} = 0. \quad (2)$$

Solutions to this equation are harmonic functions and satisfy the mean value property, meaning that the value at any interior point equals the average over its surrounding neighborhood. In diffusion problems, this corresponds to steady-state concentration fields.

B. Finite Difference Discretization

To obtain a numerical solution, the continuous domain is discretized onto a uniform square grid. Using the standard five-point stencil, the Laplace equation is approximated by

$$c_{i,j} = \frac{1}{4} (c_{i+1,j} + c_{i-1,j} + c_{i,j+1} + c_{i,j-1}), \quad (3)$$

which transforms the partial differential equation into a linear algebraic system defined at each grid point.

C. Successive Over-Relaxation

The resulting linear system is solved iteratively using the Successive Over-Relaxation (SOR) method. The update rule is given by

$$c_{i,j}^{\text{new}} = (1-\omega)c_{i,j}^{\text{old}} + \frac{\omega}{4} (c_{i+1,j} + c_{i-1,j} + c_{i,j+1} + c_{i,j-1}), \quad (4)$$

where $\omega \in (1, 2)$ is the relaxation parameter. Values $\omega > 1$ accelerate convergence relative to standard Gauss-Seidel iteration. The iteration is continued until convergence is achieved, defined by

$$\max_{i,j} |c_{i,j}^{\text{new}} - c_{i,j}^{\text{old}}| < \text{tol}. \quad (5)$$

D. Harmonic Measure and DLA

In diffusion-limited aggregation, growth occurs preferentially at boundary sites where the diffusive flux is highest. Since flux is proportional to the gradient of the concentration field, the probability of growth at a boundary site (i, j) is taken to be

$$p(i, j) \propto |\nabla c(i, j)|^\eta. \quad (6)$$

The parameter η controls the degree of nonlinear amplification. For $\eta = 1$, growth follows the harmonic measure. Larger values of η enhance tip instability and lead to thinner, more branched structures, while smaller values promote more compact growth. This coupling between diffusion and evolving geometry gives rise to screening effects and fractal morphologies characteristic of Laplacian growth.

III. METHODS

All simulations were performed on a uniform square grid of size $N = 100$. The Laplace equation was solved iteratively using the Successive Over-Relaxation (SOR) method with relaxation parameter $\omega = 1.8$. Convergence was achieved when the maximum absolute update between successive iterations satisfied $\max_{i,j} |c_{i,j}^{\text{new}} - c_{i,j}^{\text{old}}| < 10^{-6}$.

Dirichlet boundary conditions were imposed by fixing the concentration to $c = 1$ along the top boundary and $c = 0$ along the remaining boundaries. To validate the solver, an internal square region at the center of the domain was fixed to zero concentration, representing an absorbing object. The harmonic field was recomputed until convergence for each configuration.

For the diffusion-limited aggregation simulations, a single seed particle was initially placed at the center of the domain. At each growth step, the Laplace equation was solved for the current geometry. Empty lattice sites adjacent to the aggregate were identified, and the gradient magnitude of the concentration field was evaluated at those sites using central finite differences. Growth probabilities were assigned according to $p(i, j) \propto |\nabla c(i, j)|^\eta$, and one site was selected randomly using normalized probabilities. The selected site was then added to the aggregate.

For each value $\eta \in \{0.5, 1.0, 2.0\}$, 300 growth steps were performed. All simulations were implemented in Python using NumPy for numerical computations and Matplotlib for visualization.

IV. RESULTS

A. Laplace Solver Verification

The SOR solver was first tested on a square domain with fixed boundary conditions. The top boundary was set to $c = 1$, while the remaining boundaries were fixed at $c = 0$. The numerical solution converged rapidly for relaxation parameters $\omega \in [1.5, 1.9]$, with the maximum update falling below the prescribed tolerance within a few hundred iterations.

The resulting concentration field exhibits the expected harmonic behavior, with smooth monotonic decay from the top boundary toward the bottom. When an internal square sink was introduced by fixing a central region to zero concentration, the equipotential lines bend around the absorbing region, confirming correct implementation of internal Dirichlet boundary conditions.

B. Harmonic-Measure DLA Growth

Starting from a single seed particle at the center of the domain, the aggregate was grown by iteratively

solving Laplace's equation and selecting boundary sites according to the harmonic measure.

For $\eta = 1$, the cluster develops a branched fractal structure. Growth occurs predominantly at exposed tips, while interior regions become screened from the diffusive flux. This behavior is characteristic of classical DLA and demonstrates correct coupling between the Laplace field and boundary evolution.

C. Influence of the Nonlinear Parameter η

To investigate the effect of nonlinear growth amplification, simulations were performed for $\eta = 0.5$, $\eta = 1.0$, and $\eta = 2.0$ (see Figure 1). For $\eta = 0.5$, growth

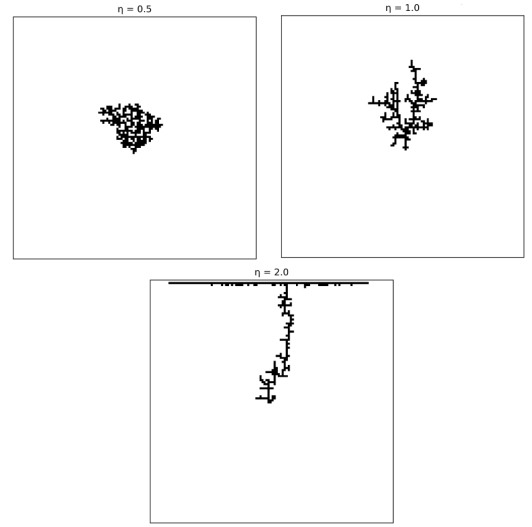


Fig. 1: Comparison of DLA clusters for $\eta = 0.5$, $\eta = 1.0$, and $\eta = 2.0$. Increasing η enhances tip amplification and leads to thinner, more unstable growth.

is more uniform and the resulting cluster is relatively compact. The reduced nonlinear amplification weakens tip selection and allows growth to occur more evenly along the boundary.

For $\eta = 1.0$, the aggregate exhibits the classical fractal morphology associated with harmonic measure, featuring multiple branching arms and clear screening effects.

For $\eta = 2.0$, the growth becomes strongly tip-dominated. The aggregate develops thin, needle-like structures with pronounced instability. Screening is enhanced, and growth concentrates almost exclusively at the most exposed regions.

These results confirm the expected transition from compact to highly unstable Laplacian growth as η increases.

V. DISCUSSION

The numerical results confirm that the SOR method provides a stable and efficient solver for the Laplace

equation under Dirichlet boundary conditions. The introduction of an internal absorbing region demonstrates correct handling of fixed interior constraints and validates the implementation.

The DLA simulations reproduce the characteristic features of Laplacian growth, including tip amplification and screening. Growth occurs preferentially at exposed boundary regions where the diffusive flux is maximal, while interior regions become progressively shielded.

The parameter η controls the nonlinear amplification of growth probability. For $\eta < 1$, growth becomes more uniform and compact. For $\eta > 1$, tip selection is enhanced, leading to highly unstable, needle-like morphologies. This transition illustrates the mechanism underlying Laplacian instability.

Limitations of the model include finite grid effects, directional bias introduced by boundary conditions, and computational cost due to repeated Laplace solves. Nevertheless, the simulations successfully capture the essential physics of diffusion-limited aggregation.

VI. CONCLUSION

In this work, a two-dimensional Laplace solver based on the SOR method was implemented and validated. The solver was coupled to a harmonic-measure growth algorithm to simulate diffusion-limited aggregation.

The simulations demonstrate how nonlinear amplification of diffusive flux leads to fractal branching and Laplacian instability. By varying the growth parameter η , a transition from compact to highly unstable morphologies was observed.

These results highlight the interplay between harmonic diffusion and boundary evolution in pattern formation.